

Number Systems [Decimal] → All base are power of 10

→ Each Digit [0, 1, 2, ... 9]

$$734 = 7 \times 100 + 3 \times 10 + 4 \times 1$$

$$6594 = 6 \times 10^3 + 5 \times 10^2 + 9 \times 10^1 + 4 \times 10^0$$

$$\begin{array}{cccc} 4 & 3 & 2 & 1 \\ \hline 10^3 & 10^2 & 10^1 & 10^0 \end{array}$$

$$N \text{ digit} = 10^N, \quad N \text{ digit} = 10^{N-1}$$

Number Systems

1) Decimal → All base value will be power of 8

2) Octal

3) Ternary

4) Binary

$$\begin{array}{cccc} (0 & 1 & 3 & 2)_8 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 8^3 & 8^2 & 8^1 & 8^0 \end{array} = 0 \times 8^3 + 1 \times 8^2 + 3 \times 8^1 + 2 \times 8^0 = 64 + 24 + 2 = 90$$

$$\begin{array}{cccc} (0 & 1 & 2 & 5)_8 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 8^3 & 8^2 & 8^1 & 8^0 \end{array} = 1 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 = 64 + 16 + 5 = 85$$

In Octal Representation: [Each Digit 0-7]

$$(10100)_8$$

$$(38745)_8$$

Ternary : base 3

$$(0 \ 2 \ 1 \ 0 \ 1) = 0 \times 3^4 + 2 \times 3^3 + 1 \times 3^2 + 0 \times 3^1 + 1 \times 3^0$$

$$\downarrow \downarrow \downarrow \downarrow \downarrow$$

$$3^4 \ 3^3 \ 3^2 \ 3^1 \ 3^0 = 54 + 1 + 1 = 64$$

In Ternary Each digit :  $[0-2]$

Binary : 2, Each Digit :  $[0, 1]$

$$(1 \ 0 \ 1 \ 1 \ 0) \Rightarrow 16 + 4 + 2 = 22$$

$$\downarrow \downarrow \downarrow \downarrow \downarrow$$

$$2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$$

Decimal  $\rightarrow$  Binary

$$\begin{array}{r} 2 \overline{) 28} - 0 \\ 2 \overline{) 14} - 0 \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ 2 \overline{) 1} - 1 \\ 0 \end{array}$$

$$(1 \ 1 \ 1 \ 0 \ 0)_2 = 28$$

$$\begin{array}{r} 2 \overline{) 35} - 1 \\ 2 \overline{) 17} - 1 \\ 2 \overline{) 8} - 0 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 2 \overline{) 1} - 1 \\ 0 \end{array} \begin{array}{l} \text{Bottom} \\ \text{up} \end{array}$$

$$(1 \ 0 \ 0 \ 0 \ 1 \ 1)_2 = 35$$

$$\begin{array}{r} 2 \overline{) 19} - 1 \\ 2 \overline{) 9} - 1 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 2 \overline{) 1} - 1 \\ 0 \end{array}$$

$$(1 \ 0 \ 0 \ 1 \ 1)_2$$

$$\begin{array}{r} 2 \overline{) 25} - 1 \\ 2 \overline{) 12} - 0 \\ 2 \overline{) 6} - 0 \\ 2 \overline{) 3} - 1 \\ 2 \overline{) 1} - 1 \\ 0 \end{array}$$

$$(1 \ 1 \ 0 \ 0 \ 1)_2$$

## Add 2 Decimal Numbers

$$\begin{array}{r}
 \begin{array}{cccc}
 1. & 1. & 1. & \\
 3 & 4 & 5 & 9 \\
 \downarrow & & & \\
 0 & 2 & 8 & 4 & 7
 \end{array} \\
 \hline
 06306
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{cccc}
 12/10 & 12/10 & 12/10 & \\
 1. & 1. & 1. & \\
 2. & 4. & 5. & 8 \\
 \downarrow & & & \\
 0 & 3. & 7. & 6. & 4
 \end{array} \\
 \hline
 06222
 \end{array}$$

$c = \text{sum}/10$   
 $d = \text{sum} \% 10$

$6/10$   
 $6\%10$   $12\%10$   $12\%10$   $12\%10$

## Add 2 Binary Numbers

$$c = \text{sum}/2, d = \text{sum} \% 2$$

Ex1:

$$\begin{array}{r}
 \begin{array}{cccc}
 1/2 & 3/2 & 2/2 & 1/2 \\
 0. & 1. & 1. & 0. \\
 1. & 0. & 1. & 1. & 0
 \end{array} \\
 \hline
 00111
 \end{array}$$

$1\%2$   $3\%2$   $2\%2$   $1\%2$

11101

Ex2:

$$\begin{array}{r}
 \begin{array}{cccc}
 2/2 & 2/2 & 3/2 & 2/2 \\
 1. & 1. & 1. & 1. \\
 3/2 & 1. & 1. & 0. & 1. & 1
 \end{array} \\
 \hline
 110010
 \end{array}$$

$3\%2$   $2\%2$   $2\%2$   $3\%2$   $2\%2$

110010<sub>2</sub>

Ex1:

$$\begin{array}{r}
 \begin{array}{cccc}
 1. & 1. & 1. & 1. \\
 1. & 0 & 1. & 1. & 0. \\
 1 & 1. & 0 & 1. & 1. \\
 1 & 0 & 1. & 0 & 1.
 \end{array} \\
 \hline
 1000110
 \end{array}$$

$c = 4/2$   
 $2/2$   
 $2$   
 $2\%2$   $4\%2$   $3\%2$   $2\%2$

$32 + 16 + 4 + 2 = 54$

$22$   
 $11$   
 $21$

54

## Naming Conventions

1 → Set  
0 → unset

Ex:  $\begin{array}{cccccc} \text{6} & \text{5} & \text{4} & \text{3} & \text{2} & \text{1} & \text{0} \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{array}$  }  
 Is 3 bit Set: YES  
 Is 4 bit Set: NO

$[1, N]$  : N Numbers

$[0, N-1]$  = N Numbers

$[0, N]$  : N+1 Numbers

$[0, N-2]$  = N-1 Numbers

$$[a, b] = b - a + 1$$

## Geop/Geometric Progression

$$2^0 + 2^1 = 3 (2^2 - 1)$$

$$2^0 + 2^1 + 2^2 = 7 (2^3 - 1)$$

$$2^0 + 2^1 + 2^2 + 2^3 = 15 (2^4 - 1)$$

$$2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^{N-1} = 2^N - 1$$

$$2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^{N-1}$$

Sum of T terms in Geop

$$\frac{a(r^t - 1)}{r - 1} \quad \left\{ \begin{array}{l} a = 1^{\text{st}} \text{ term value} \\ r = \text{common ratio} \end{array} \right.$$

$$a = 1, t = N, r = 2$$

$$\frac{1(2^N - 1)}{2 - 1} = 2^N - 1$$

MSB - Most Significant Bit / Left Most Bit

LSB - Least Significant Bit / Right Most Bit

8 bit binary Number : 10

$$\begin{array}{ccccccc} & & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{array}$$

8 bit binary Number : -10

$$\{1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0\}$$

$$\begin{array}{ccccccc} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ - & - & - & - & - & - & - & - \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ -2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{array}$$

$$1 \times (-2^7) + 1 \times (2^3) + (1) \times 2^1$$

$$\rightarrow -128 + 8 + 2 = -118$$

Ex2:

8 bit :

$$\begin{array}{ccccccc} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ - & - & - & - & - & - & - & - \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ -2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{array}$$

$$-128 + 32 + 4 + 1 = -91$$

4 bit Number

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ - & - & - & - \\ 3 & 2 & 1 & 0 \\ -2 & 2 & 2 & 2 \end{array} \} = -7$$

8 bit :

$$\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$16 + 4 + 1 = 21$$

6 bit Number

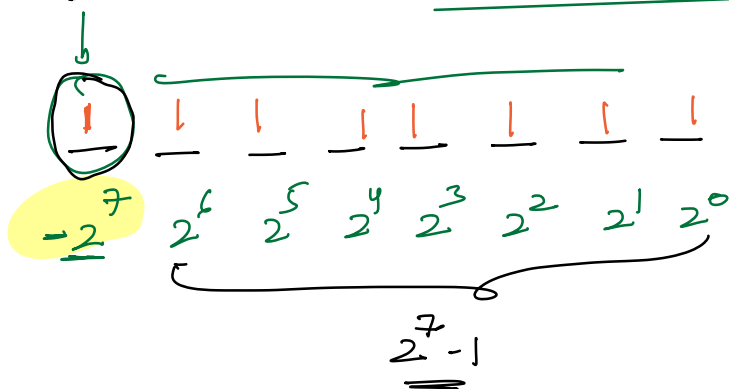
$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 1 & 1 \\ - & - & - & - & - & - \\ 5 & 4 & 3 & 2 & 1 & 0 \\ -2 & 2 & 2 & 2 & 2 & 2 \end{array} \} \begin{array}{l} -32 + 8 + 2 \\ + 1 \\ = -21 \end{array}$$

$$\text{In } N \text{ bit Number MSB} = -2^{N-1}$$

→ Computer Store Negative Numbers as 2's

8 bit Number [MSB value can dominate all other bits]

combined



$$\begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \downarrow & & & & & & & \\ -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array} = -128$$

4 bit Number

$$\begin{array}{ccc} 1 & 1 & 1 \\ \downarrow & \downarrow & \downarrow \\ -2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

Can we store +10 here =  $\times$  Max 7

Can we store -10 here =  $\times$  Min -8

| # bits | Min                                                                            | Max                                                                           |
|--------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 2      | $\frac{1}{-2^1} \frac{0}{2^0} \} = -2$                                         | $\frac{0}{-2^1} \frac{1}{2^0} = 1$                                            |
| 3      | $\frac{1}{-2^2} \frac{0}{2^1} \frac{0}{2^0} = -4$                              | $\frac{0}{-2^2} \frac{1}{2^1} \frac{1}{2^0} = 3$                              |
| 4      | $\frac{1}{-2^3} \frac{0}{2^2} \frac{0}{2^1} \frac{0}{2^0} = -8$                | $\frac{0}{-2^3} \frac{1}{2^2} \frac{1}{2^1} \frac{1}{2^0} = 7$                |
| 5      | $\frac{1}{-2^4} \frac{0}{2^3} \frac{0}{2^2} \frac{0}{2^1} \frac{0}{2^0} = -16$ | $\frac{0}{-2^4} \frac{1}{2^3} \frac{1}{2^2} \frac{1}{2^1} \frac{1}{2^0} = 15$ |

N bits

$$\left\{ \frac{1}{-2^{N-1}} \frac{0}{2^2} \frac{0}{2^1} \frac{0}{2^0} \dots \right\} = \frac{N-1}{-2} \quad \text{Min}$$

$$\left\{ \frac{0}{-2^{N-1}} \frac{1}{2^{N-2}} \frac{1}{2^{N-3}} \dots \frac{1}{2^2} \frac{1}{2^1} \frac{1}{2^0} \right\} = \frac{N-1}{2} - 1 \quad \text{Max}$$

$$2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^{N-2} = \frac{2^{N-1}}{2} - 1 \quad \left\{ \frac{N-1}{-2}, \frac{N-1}{2} - 1 \right\}$$

byte : 8 bit Number  $\Rightarrow \{-2^7, 2^7-1\} \Rightarrow \{-128, 127\}$   
1 byte

short : 2 bytes  $\rightarrow 16$  bits  $\Rightarrow \{-2^{15}, 2^{15}-1\} \Rightarrow \{-32768, 32767\}$

int : 4 bytes  $\rightarrow 32$  bits  $\Rightarrow \{-2^{31}, 2^{31}-1\} \approx [-2 \times 10^9, 2 \times 10^9]$

$\left[ \underbrace{-2,147,483,648}_{\text{INT\_MIN}}, \underbrace{2,147,483,647}_{\text{INT\_MAX}} \right]$   $\begin{matrix} \text{arr[0]} \\ 1 <= 10^9 \end{matrix}$

long : 8 bytes  $\rightarrow 64$  bits  $\Rightarrow \{-2^{63}, 2^{63}-1\} \approx [-8 \times 10^{18}, 8 \times 10^{18}]$

### Approximation

$$2^{10} = 1024 \approx 1000 = 10^3$$

$$\underline{2^{10} \approx 10^3}$$

Apply whole 2 on both sides  $\rightarrow$

$$(2^{10})^3 \approx (10^3)^3 \Rightarrow \underline{2^{30} \approx 10^9} \xrightarrow{\text{Mult by 2}} \underline{2^{31} \approx 2 \times 10^9}$$

$$(2^{10})^6 \approx (10^3)^6 \Rightarrow \underline{2^{60} \approx 10^{18}} \xrightarrow{\text{Mult by 8}} \underline{2^{63} \approx 8 \times 10^{18}}$$

10:50 PM break

Bit Wise operators : ( &, |, ^, ~, <<, >> ) → Tomorrow Session

Truth Table

same: 0 / diff: 1

| a     | b | a & b | a   b | a ^ b | ~a |
|-------|---|-------|-------|-------|----|
| 0 - 0 |   | 0     | 0     | 0     | 1  |
| 0 - 1 |   | 0     | 1     | 1     | 1  |
| 1 - 0 |   | 0     | 1     | 1     | 0  |
| 1 - 1 |   | 1     | 1     | 0     | 0  |

byte a = 29 :  $\begin{matrix} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \hline 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{matrix}$

byte b = 18 :  $\begin{matrix} \hline 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{matrix}$

Decimal

print (a & b) : 0 0 0 1 0 0 0 0 : 16

print (a | b) : 0 0 0 1 1 1 1 1 : 31

print (a ^ b) : 0 0 0 0 1 1 1 1 : 15

print (~a) : 1 1 1 0 0 0 1 0

$$-128 + 64 + 32 + 2 = -128 + 98 = -30$$

byte a = 13 : 0 0 0 0 1 1 0 1

byte b = 10 : 0 0 0 0 1 0 1 0

print(a & b) : 0 0 0 0 1 0 0 0 : 8

print(a | b) : 0 0 0 0 1 1 1 1 : 15

print(a ^ b) : 0 0 0 0 0 1 1 1 : 7

Decimal

Property:

byte a = 10 LSB = 0

a = 0 1 0 1 0 0 0 0

1 : 0 0 0 0 0 0 0 1

a & 1 : 0 0 0 0 0 0 0 0

if (a & 1 == 0) {

LSB of a = 0

a is even

}

// a & 0 = \_\_\_\_\_

byte a = 11

a = 0 1 0 1 1 0 0 1

1 = 0 0 0 0 0 0 0 1

if (a & 1 == 1) {

LSB of a = 1

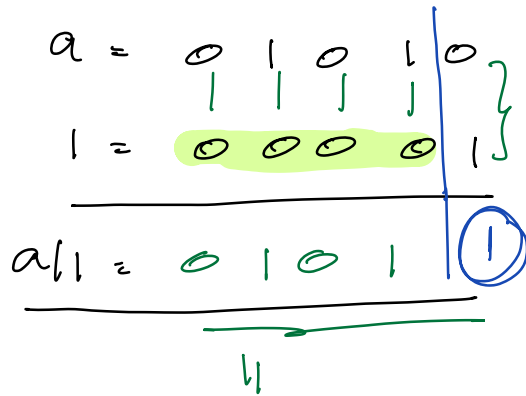
a is odd

}

a & a = \_\_\_\_\_

odd value  
= 1

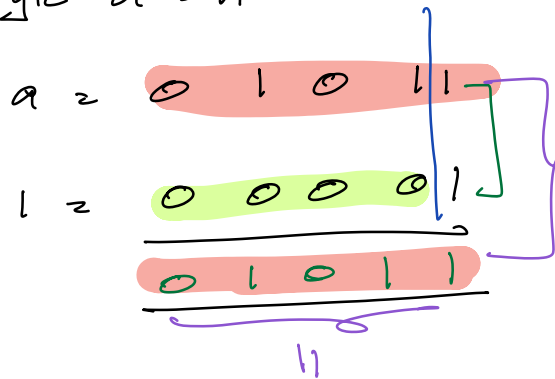
byte  $a = 10$



$$a|l = (a+1)$$

$a$  is even

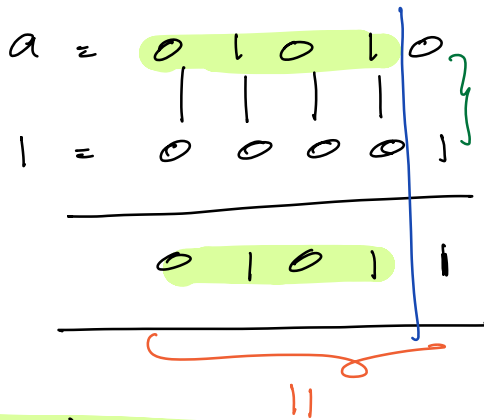
byte  $a = 11$



$$a|l = (a)$$

$a$  is odd

byte  $a = 10$



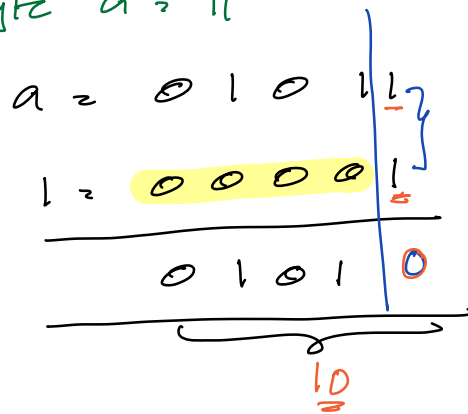
$$(a^l) = a+1$$

$a$  is even

$$a \wedge a = 0$$

$$a^0 = a$$

byte  $a = 11$



$$(a^l) = (a-1)$$

$a$  is odd

$$a \in b = b \in a$$

$$a \mid b \Rightarrow b \mid a$$

$$a \cap b = b \cap a$$

Commutative property

$$(a \in b \in c) \rightarrow (a \in c \in b) \leftarrow$$

$$(b \in \mathcal{L} \in a) \leftarrow$$

$$\underline{(a \in (b \in c))} \rightarrow \underline{(c \in c) \wedge a}$$

$$(a \in (c \in b)) = (a \in c \in b)$$

## Associative property

$$a|b|c = a|c|b = b|c|a$$

$$a^{\wedge} b^{\wedge} c = a^{\wedge} c^{\wedge} b = b^{\wedge} c^{\wedge} a$$

observations

$$\overbrace{a^1 b^1 a} \Rightarrow \overbrace{a^1 a^1} b = \overbrace{0^1} b = b$$

$$\overbrace{a^1 c^1 a^1 b^1} \underbrace{b} = \underbrace{a^1 a^1}_0 \underbrace{b^1 b^1}_0 c = c$$

$$\cancel{a^1} \cancel{c^1} \cancel{d^1} \cancel{c^1} \cancel{d^1} c^1 \cancel{d^1} = c$$

1Q) Given N array element, every element repeats twice except 1 elem find unique element

$$arr[7] = 4 \wedge 7 \wedge 6 \wedge 4 \wedge 8 \wedge 7 \wedge 6 \Rightarrow \underline{8} \quad \text{ans}$$

$$arr[5] = 1 \wedge 4 \wedge 4 \wedge 3 \wedge 1 \Rightarrow \underline{3} \quad \text{ans}$$

Sol: XOR of all Elements

// sum of all array elements

$S = 0$

$P = 0; P < N; P++ \{$

$|$

$S = S + arr[P]$

$\}$

if we replace + with ^ that becomes xor of all Elements

Note: Bitwise operators are faster when compared to arithmetic operators.

if  $(a \% 2 == 0)$  | if  $(a \% 2 == 0)$   
a is even | a is even

Doubts:

$\&$   $\rightarrow$  bitwise operator

$\&\&$   $\rightarrow$  logical operator

int  $\rightarrow$  4B  $\rightarrow$  32bits  $\Rightarrow$  MSB  $=$   $-2^{31}$

arr[5] =  $\begin{array}{|c|c|c|c|c|} \hline 1 & 4 & 4 & 3 & 1 \\ \hline \end{array} \rightarrow$   
ans = 0  $\begin{array}{|c|c|c|c|c|} \hline 1 & 4 & 1 & 3 & 3 \\ \hline \end{array}$

8bit:  
 $\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$   
 $\rightarrow -2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

32bit  $\Rightarrow$   $-2^{31}$   
 $\rightarrow$   
int